



Oracle Database @ AWS Automation

Operations Advisory Team

Office of CTO | EMEA Technology Engineering

May 2026

Agenda

1. **Oracle Database at AWS Cloud architecture**
2. **Operations & Interfaces**
3. **IaC & GitOps**
4. **Wrap up & next steps**
5. **Q&A**

ODB@AWS

Architecture

Available Database Services



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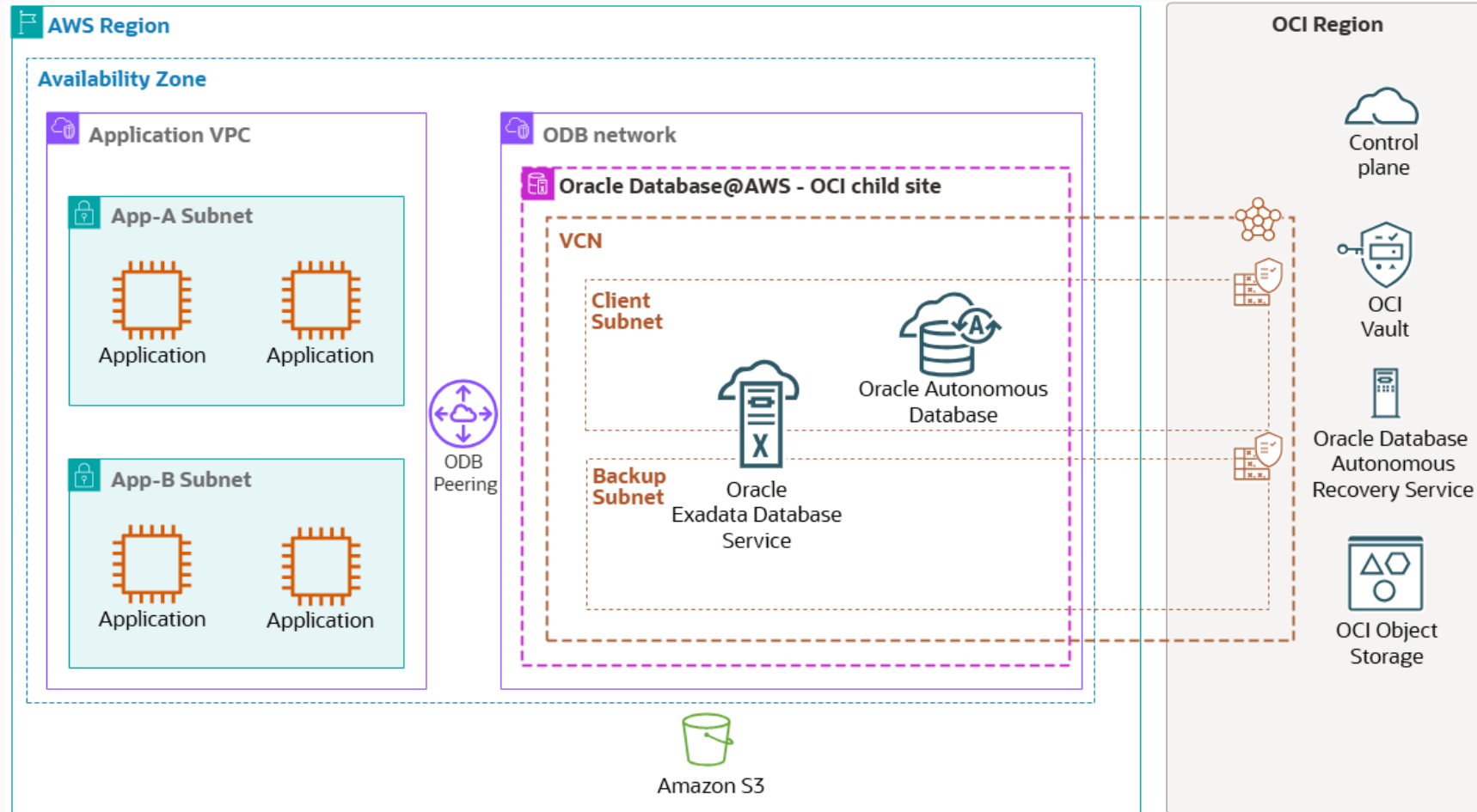
***Oracle Exadata Database
Service on Dedicated
Infrastructure***



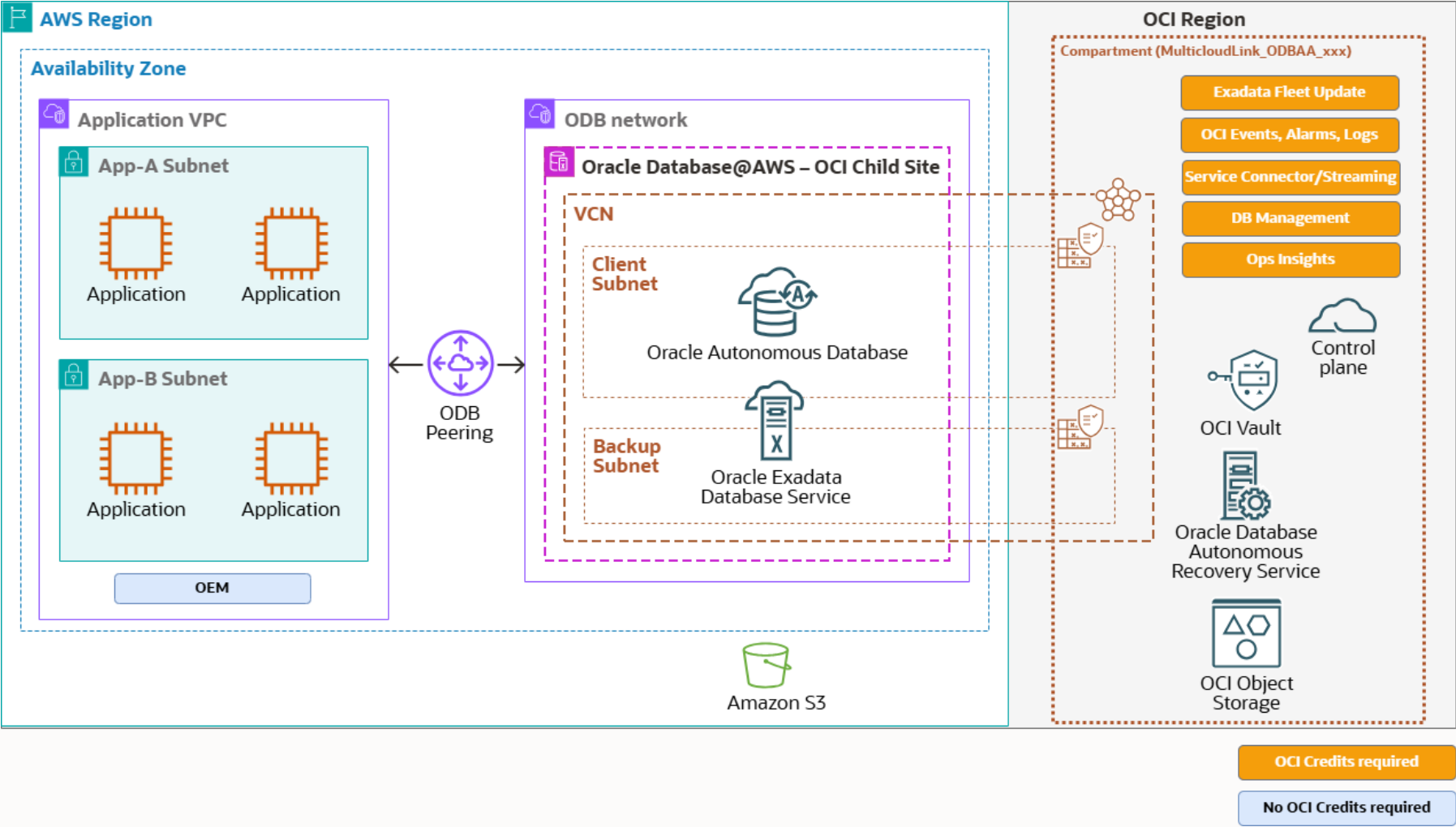
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***Oracle Autonomous AI
Database on Dedicated
Exadata Infrastructure***

Architecture overview

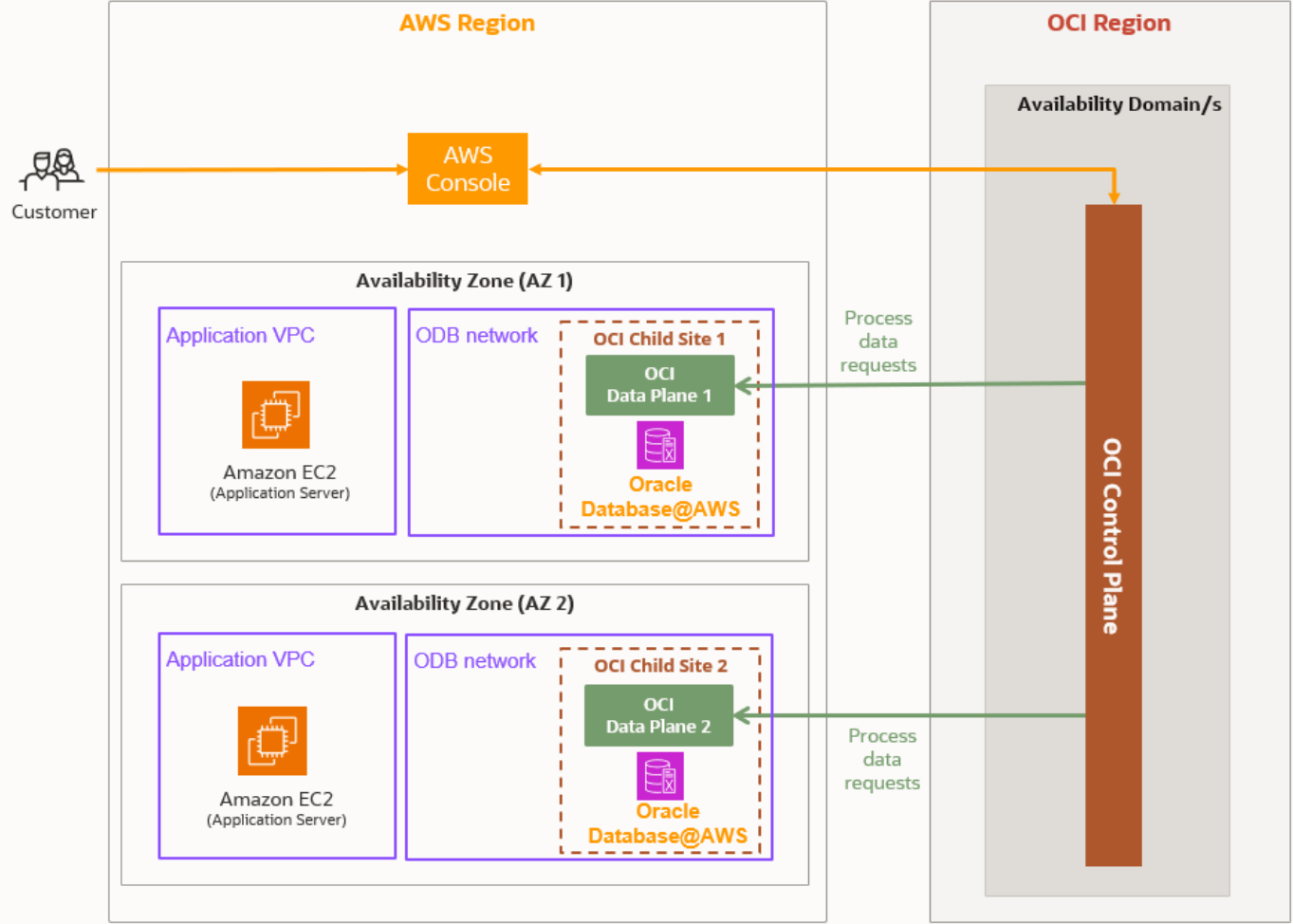


Optional operational services

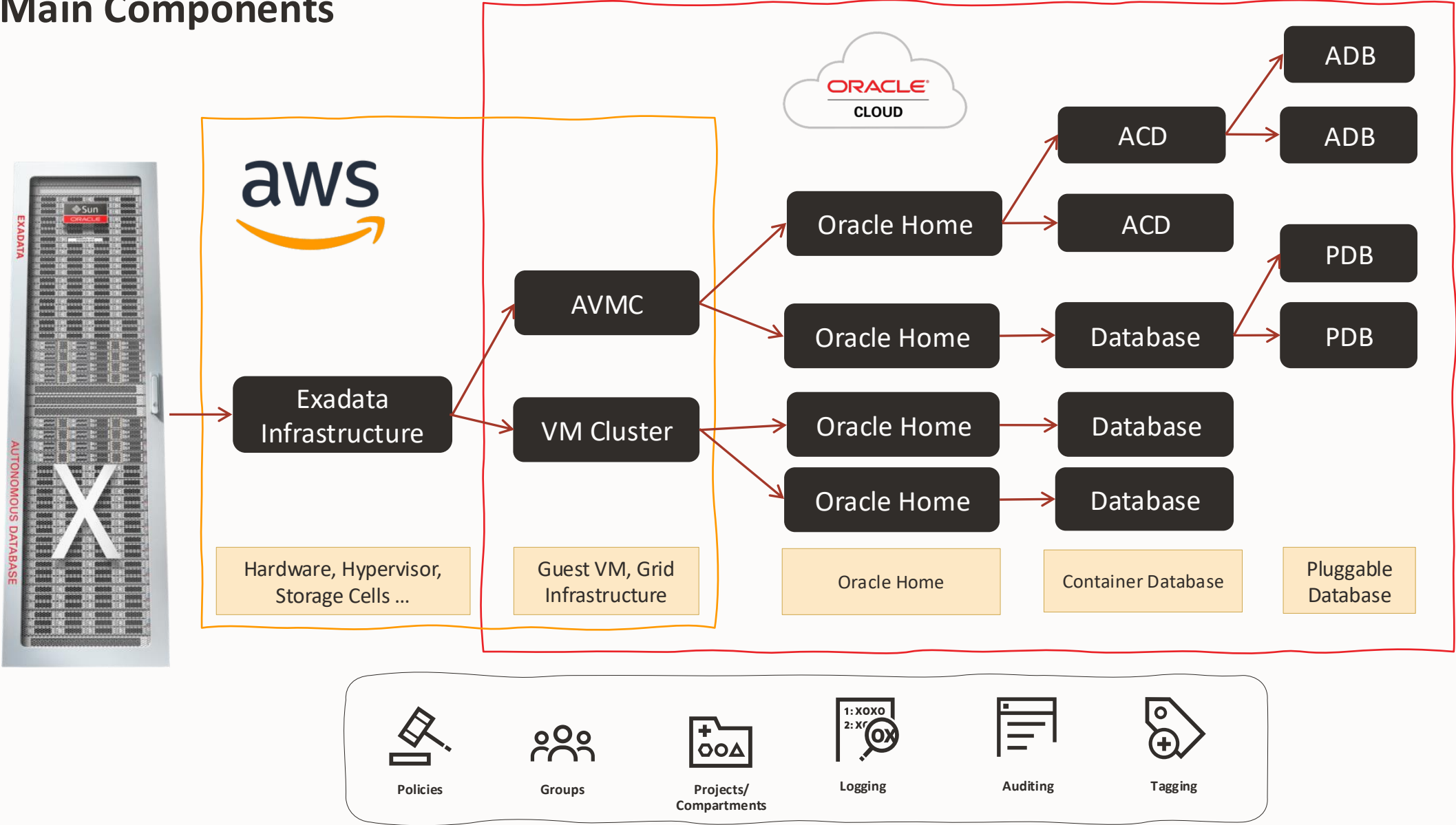


Operations & Interfaces

Control Plane Architecture



Main Components



Management Interfaces

Rest APIs

REST APIs that use HTTPS requests and responses for managing OCI and AWS resources, including Exadata Cloud

Consoles

Intuitive, graphical interfaces for creating and managing your OCI and AWS resources, are one of the interfaces to the Rest APIs

Command Line Interfaces (OCI CLI / AWS CLI)

Lightweight tools for command-line tasks. Support scripts and extend Console functionality

Software Development Kits (SDKs)

Enable custom solutions for Exadata Cloud and OCI and AWS services

DevOps Tools & Plug-ins

Tools like Terraform Provider and Ansible Collection for provisioning and automation

Cloud Shell

Browser-based terminal with pre-authenticated CLI, useful for tutorials and quick commands

Local VM Command-Line Interfaces

CLI utilities located on the VM guests

Provisioned as part of the VM clusters on the Exadata Cloud Infrastructure, are available to perform various lifecycle and administration operations

dbaascli

Use the dbaascli utility to perform various database lifecycle and administration operations on the Exadata Cloud Infrastructure such as changing the password of a database user, starting a database, managing pluggable databases (PDBs), and more.

ExaCli

Use the ExaCLI command-line utility to perform monitoring and management functions on Exadata storage servers in the Exadata Cloud.

bkup_api

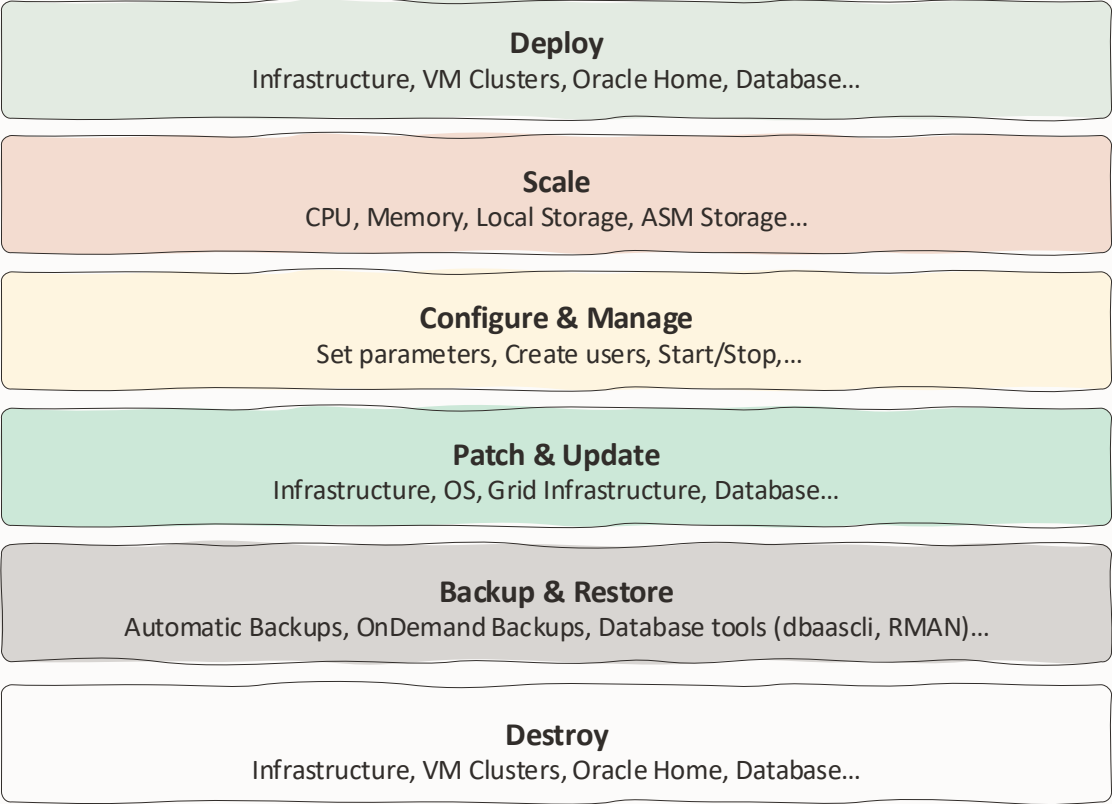
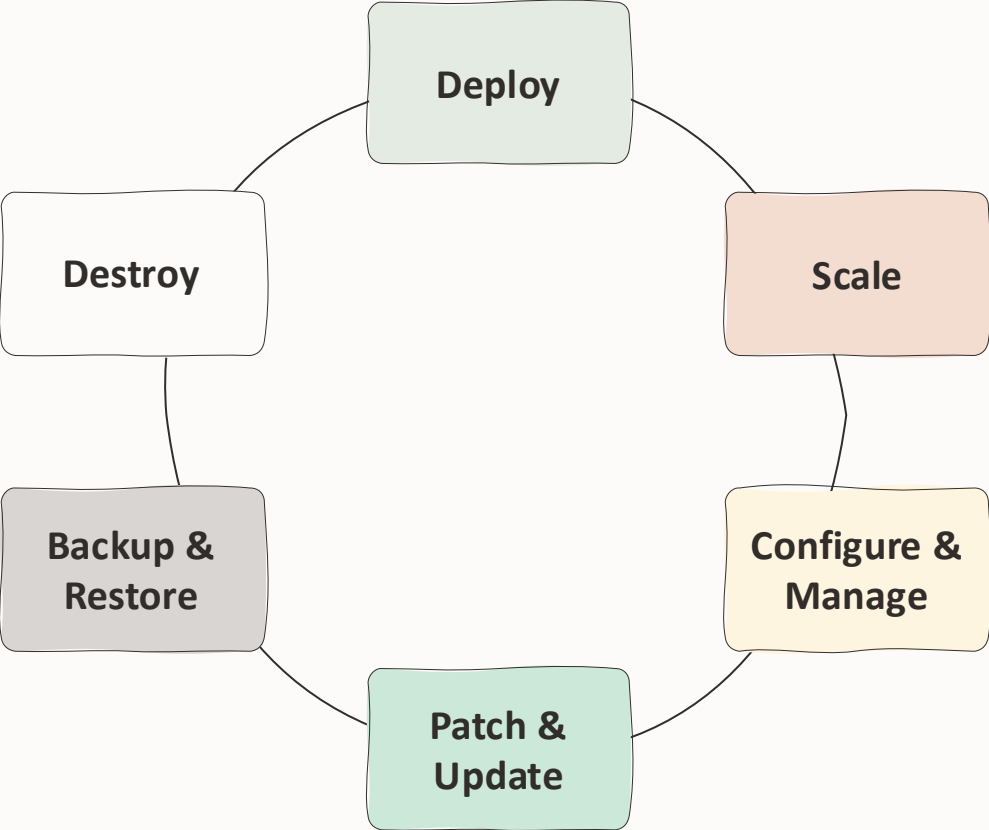
Handles backup/recovery tasks, including on-demand database and PDB backups.

DEPRECATED

Must be connected to a VM in the Exadata Cloud VM cluster;
uses VM OS user security, not OCI user security.

Changes sync to OCI Control Plane via **DB Sync**.
Tools are auto-updated by Oracle on a regular basis.

Operations



Where I do what?



- ODB Network and subnets
 - Creation/Deletion
 - View information
- Exadata infrastructure
 - Creation/deletion
 - Updates/Upgrades planning
 - View Information
- Exadata VM clusters and Autonomous VM clusters
 - VM cluster creation/deletion
 - Status management
 - View information
- ODB peering connections
- *Oracle CDBs/ACDs and PDBs/ADB aren't provisioned and managed on AWS Cloud but in OCI*



- Exadata infrastructure
 - Configuration changes
- Exadata VM clusters and Autonomous VM clusters
 - Updates/Upgrades
 - Scale OCPU/Memory/Storage
- Oracle Home creation/deletion
- Oracle CDBs/ACDs and PDBs/ADBs
 - Creation/Deletion
 - Status management
 - Updates/Upgrades
 - Backup & Recovery
 - DR setup and management
 - (...)

Management Interfaces vs. Operations

Operation	Amazon Cloud				OCI					IaC	
	Console	AWS CLI	Rest API	SDK	Console	OCI CLI	dbaascli	Rest API	SDK	Terraform ¹	Ansible Collections
Create/Delete ODB Network and ODB Subnets	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Create/Delete Exadata Infrastructure instances	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Scale Exadata Infrastructure instances	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✓
Create/Delete Exadata VM Cluster	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Stop/Start/Restart VM clusters	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✓
Create/Delete Database Oracle Homes	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Create/Delete Container Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Stop/Start/Restart Container Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Create/Delete/Clone Pluggable Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Stop/Start/Restart Pluggable Databases	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Patch Guest OS/GI	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✓
Patch CDB/PDB	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Database Backup&Restore	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓
Scale VM OCPU/Memory/Storage (Local/Shared)	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✓
Schedule Infrastructure Maintenance	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗
Monthly Security Maintenance	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
Create software images	✗	✗	✗	✗	✓	✓	✗	✓	✓	✓	✓
Create/Delete Data Guard Association	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓
Perform Switchover/Failover	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓

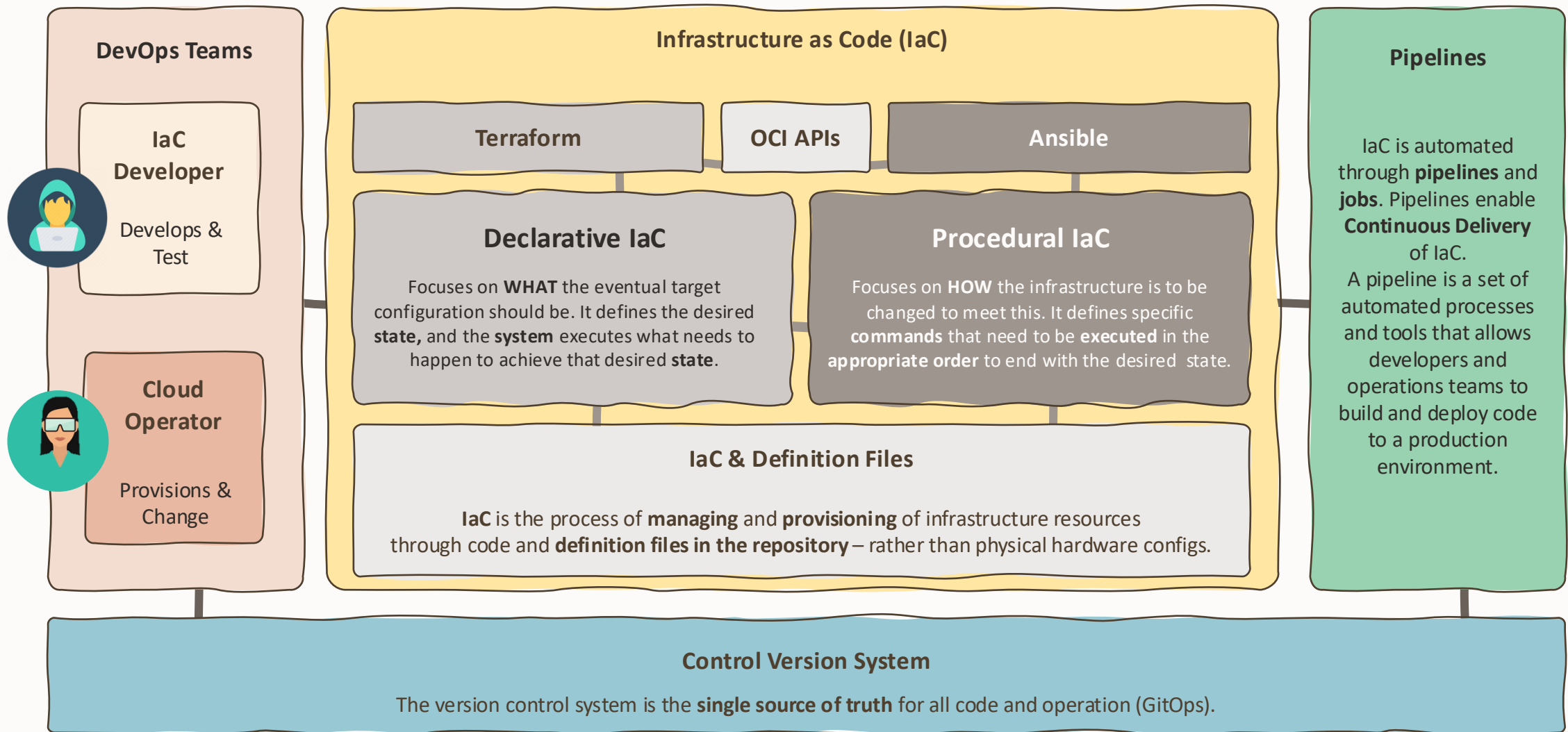


laC & GitOps

What is IaC?

*Infrastructure as Code is the process of **provisioning and managing** computing infrastructure through **code** instead of manual processes.*

IaC Technology Context & Personas



What is and does Terraform

Terraform is a **declarative** language that defines the **desired state** of infrastructure.

It uses a **state file** to consistently provision **Landing Zones (LZ)** and **workload** resources across **any cloud**.

Terraform is great to **provision the Landing Zone and workload** resources.



Declarative Language

Defines the "**what**", not the "how". A simple configuration expresses the entire desired state.



State Management

A **state file** acts as the **single source of truth**, mapping configurations to real-world resources.



Landing Zones Workload Resources

LZ set the **foundation** and **guardrails**.
Workload resources add the stacks on top across **any cloud**.

OCI Terraform Provider Resources



ODB Network
[aws_odb_network](#)



Exadata Infrastructure
[aws_odb_cloud_exadata_infrastructure](#)



VM Cluster
[aws_odb_cloud_vm_cluster](#)



Autonomous VM Cluster
[aws_odb_cloud_autonomous_vm_cluster](#)



VM Cluster
[oci_database_cloud_vm_cluster](#)



DB Home
[oci_database_db_home](#)



CDB
[oci_database_database](#)



PDB
[oci_database_pluggable_database](#)



Terraform is not enough



The Limits

Terraform has some boundaries. It cannot manage tasks inside a resource, only the resource itself.

-  **No patching:** Not for OS/DB/app patching or upgrades
-  **No in-guest config:** Doesn't install packages, edit files, or manage services on existing hosts
-  **No app/data config:** doesn't manage DB schemas/users or app settings.
-  **Non-Updatable** changes (needs to re-deploy)
-  **Last resort only:** Provisioners (remote-exec/local-exec) and generic helpers (terraform_data, deprecated null_resource) aren't suited for Config Management and can leak secrets to state/logs

The Solution

Use the [lifecycle statement](#) block to **bridge the gap** between **declarative IaC** and **procedural** realities.

```
lifecycle {
  ignore_changes = [
    database[0].admin_password,      <--- Ignoring by default
    database[0].database_software_image_id <--- ignore db_home patching
    database_software_image_id,      <--- ignore db_home patching
    db_version,                      <--- ignore db_home patching
  ]
}
```

The [ignore_changes](#) skip diffs for specific fields (config edits or drift)

What is Ansible

Ansible is a declarative, **YAML-based** automation engine, **agentless** IaC tool used primarily for **configuration management**.

Agent less

No permanent **agents** on managed systems.
Uses **temporary** network **connections** via **SSH** or
Windows Remote Management

Vs. Alternatives

Chef, Puppet and Salt require installed agents.
Ansible eliminates this **complexity** and overhead

Minimum Requirement

Only requires Python installed on managed
systems.
Nothing more

Central Orchestration

Centralized and efficient **architecture**
All **orchestration** is done from a **control node**
(masterd host).

SSH Security

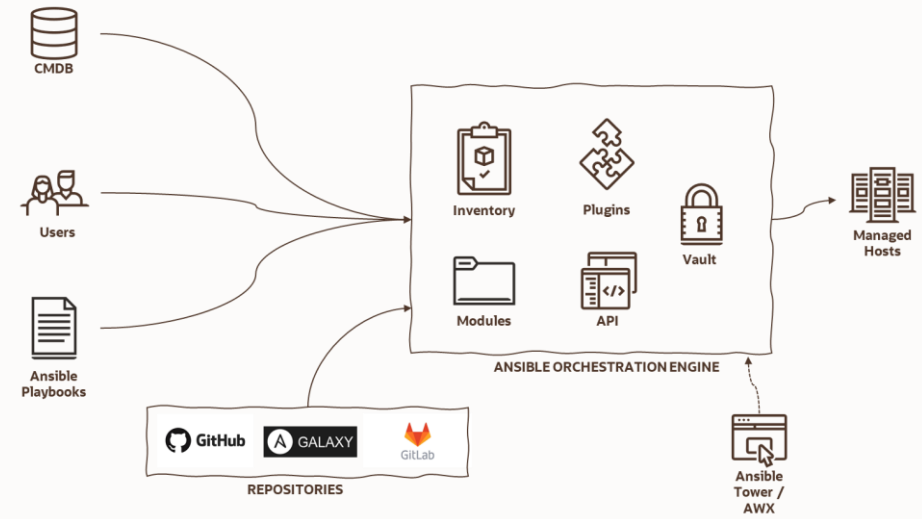
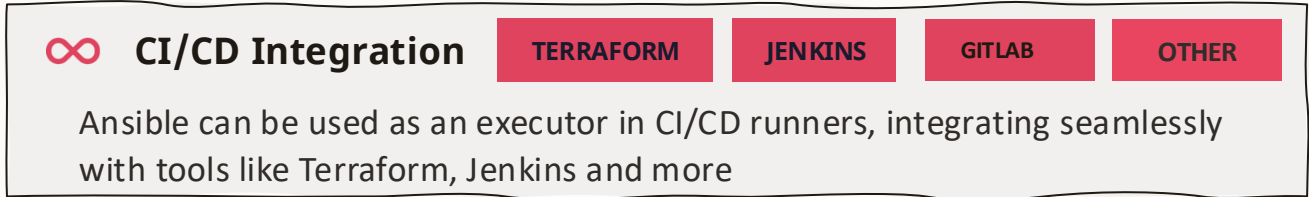
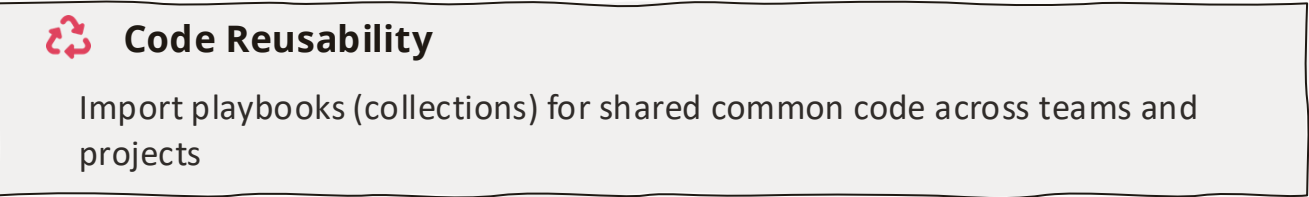
Leverages native **SSH security** without additional
ports or running daemons

Zero Overhead

Zero resource consumption on managed servers
when no tasks are running

Playbooks: The Heart of Ansible

Playbooks are the language by which Ansible orchestrates, configures, administers, or deploys systems.



OCI Ansible collection

Ansible Collection helps you **automate the provisioning and configuring** of Oracle Cloud Infrastructure services and resources

Github repository

[OCI Ansible collection](#)

Ansible Tower/ AWX Integration

[Ansible Tower](#)

[AWX.](#)

Ansible Inventory Plugin

[Static Inventory](#) & [Dynamic inventory](#)

Supported Services

The Oracle Cloud Infrastructure Ansible Collection provides **hundreds of [modules](#)** to **automate** provisioning and configuration of **OCI resources**.

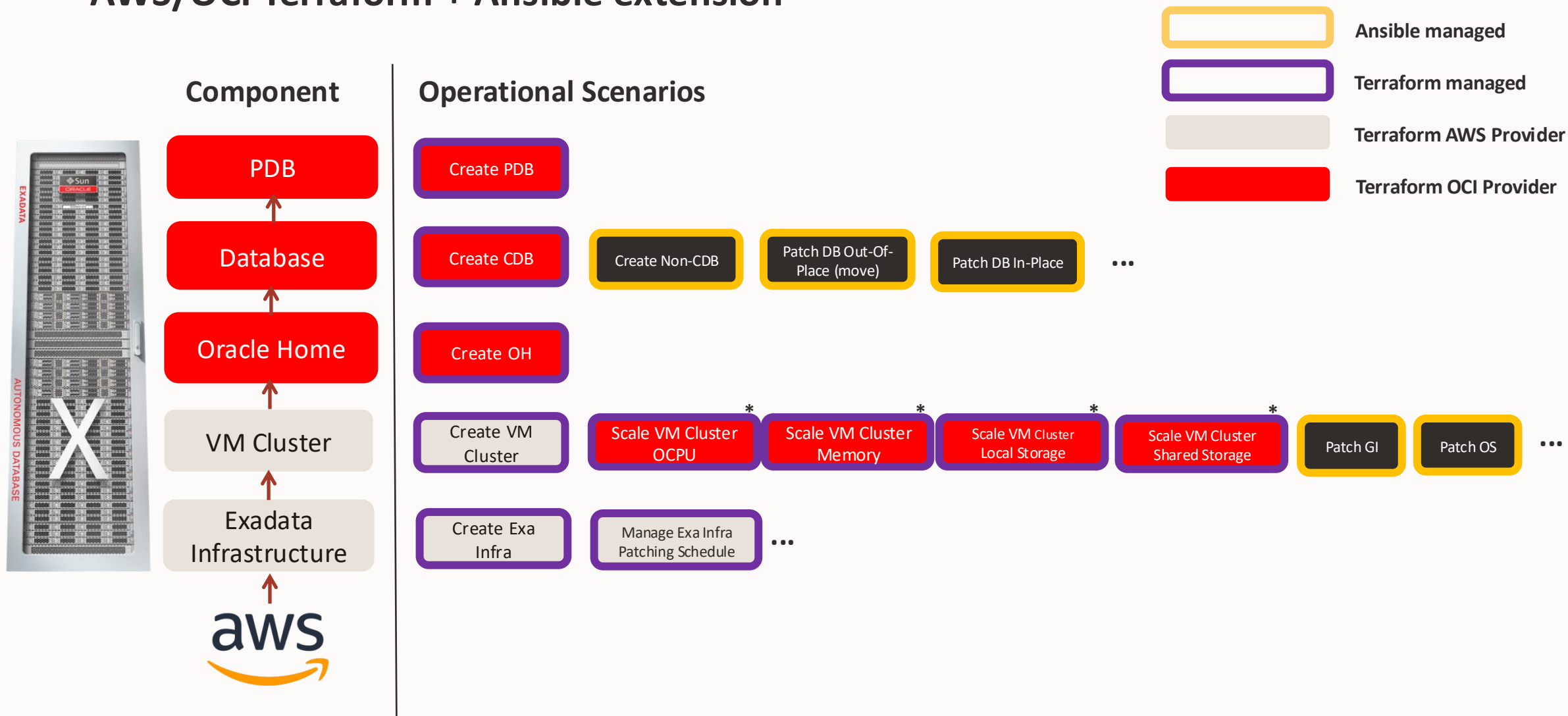
Minimal Requirements

The OCI **Ansible collection** requires Ansible version. **2.9 or later**

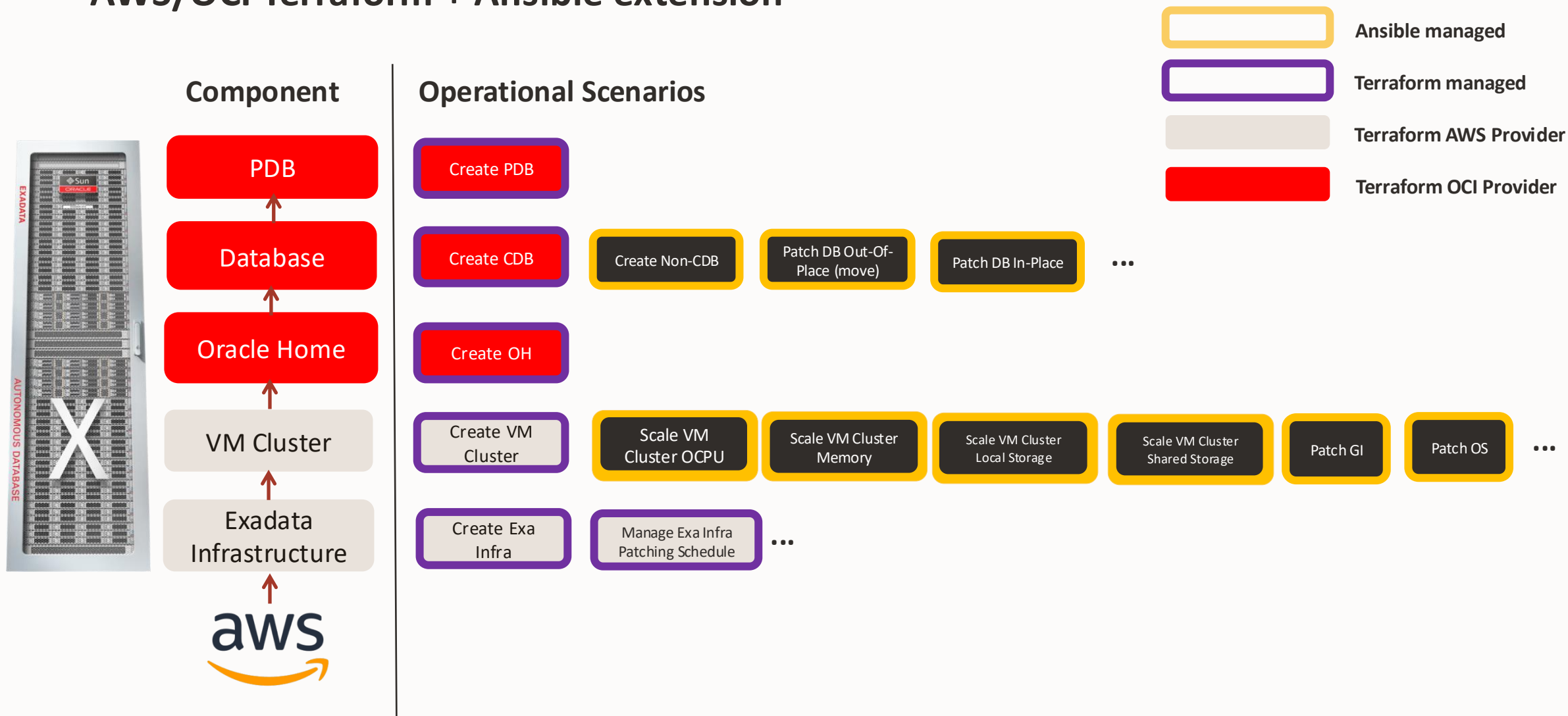
OCI Collection key benefits

Provision & configure OCI via Ansible.
Dynamic inventory for discovery.
Centralized runs (AWX/Tower, OLAM).
Broad, Oracle-maintained coverage.

AWS/OCI Terraform + Ansible extension



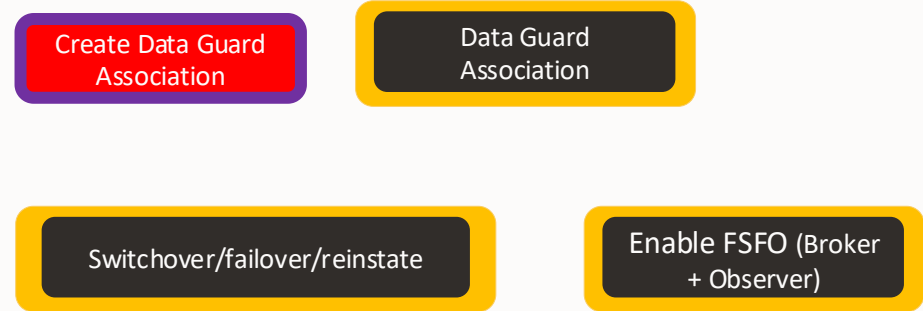
AWS/OCI Terraform + Ansible extension



G+D Operations



Data Guard & Fast-Start Failover (FSFO)



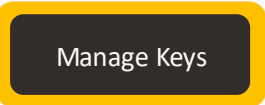
Unified Audit & Data Safe



Database Backup & Restore



Oracle Key Vault



OCI Vault

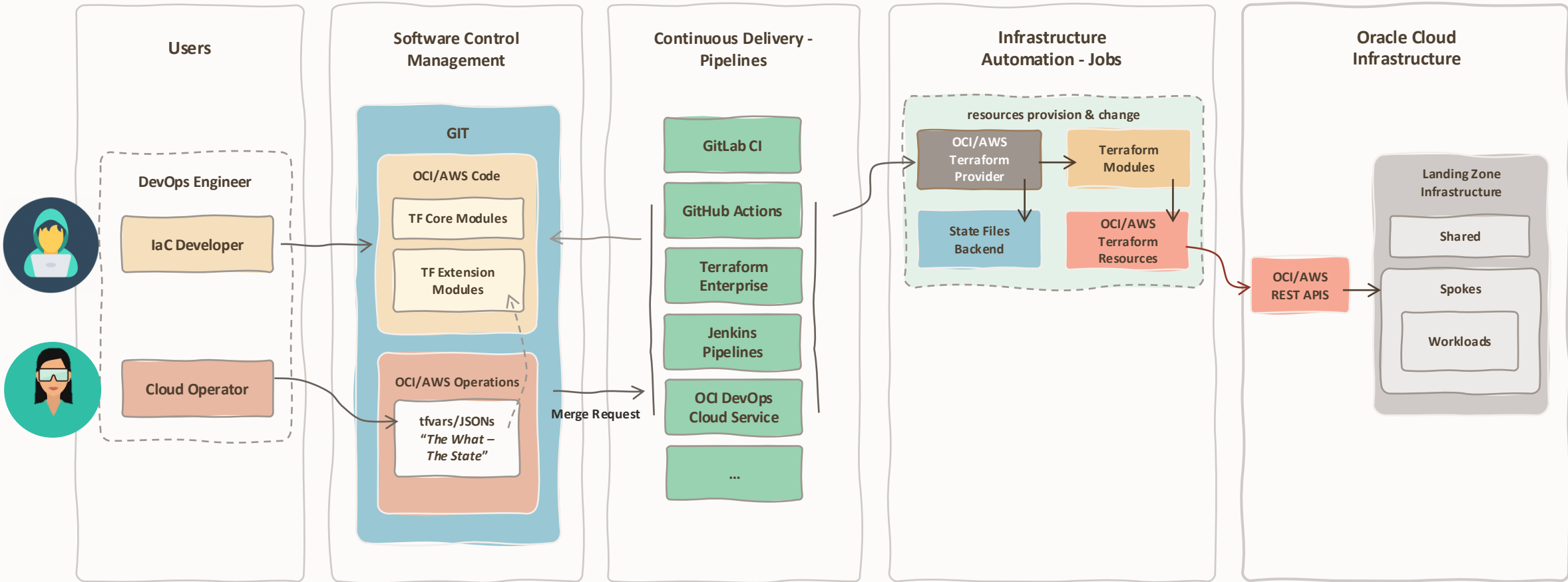
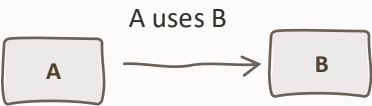


What is GitOps?

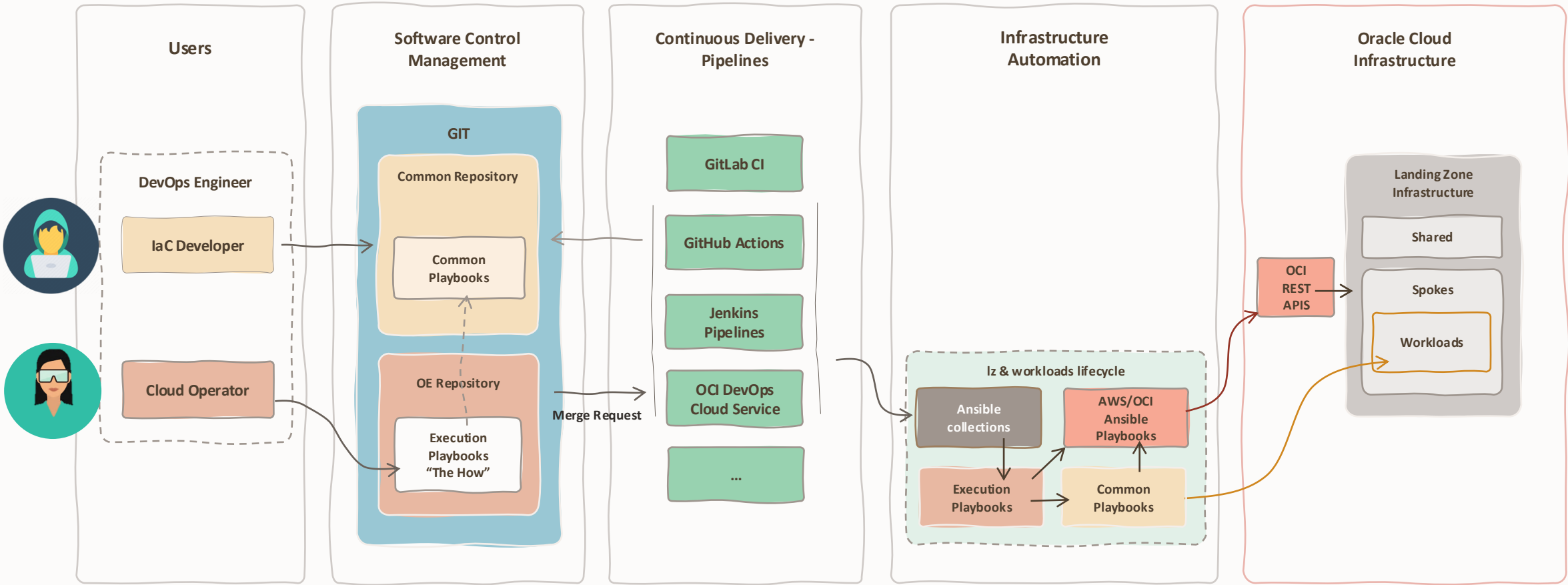
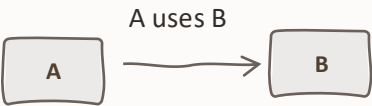
*GitOps is an operational practice that **applies DevOps best practices**, like **version control, collaboration, compliance**, and **CI/CD** to infrastructure automation.*

GitOps becomes a great common multi-cloud operating model.

GitOps using a Declarative Pattern with Terraform



GitOps using a Procedural Pattern with Ansible



Wrap up & Next Steps

Takeaways | Next steps

- There is ***not a single tool to manage everything.***
- Different management interfaces to manage ExaDB-D components, ***leverage to your preferred automation tooling.***
- The use of a ***GitOps*** approach ***simplifies your operations.***
- ***Exadata Fleet Update facilitates*** the schedule, patching process and automation needed.
- OCI Basic monitoring & integration services can be used to forward valuable information to your IT Systems (SIEM, ITSM, etc.).

Additional Operations Workshops

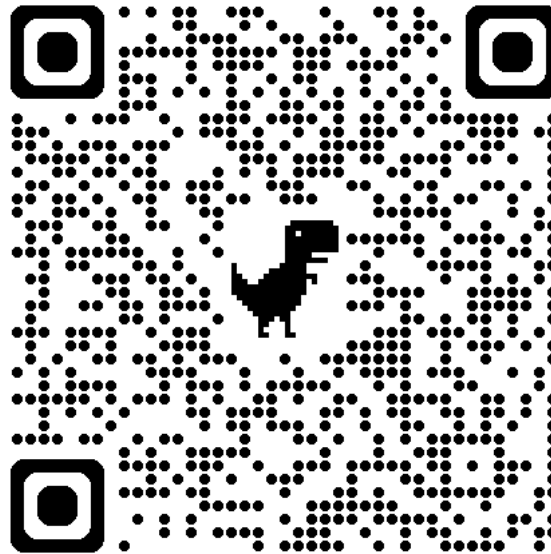
Exadata Fleet Update

Learn what is Exadata Fleet Update and how it can be used to patch Exadata OS, GI & DBs at scale

OD@ExaDB-D Operations Deep Dive

Learn how you can use different management interfaces to provision, change, maintain & manage high availability for Exadata Cloud Service in AWS

Stay tuned



[Operations Advisory](#)
[GitHub Repository](#)

Thank you

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